

Stats 21 Practice Final — Answer Key

Every answer below was verified by running the exact code in Python (numpy 2.4 / pandas 3.0). Q19 is blank in the practice PDF (no code was shown), so it is skipped. 49 questions total.

Tip: grade yourself with the quick grid first, then read the explanations only for the ones you missed.

Quick answer grid

Q1: a	Q2: a	Q3: b	Q4: b	Q5: b	Q6: b	Q7: b
Q8: c	Q9: c	Q10: a	Q11: c	Q12: a	Q13: b	Q14: b
Q15: b	Q16: b	Q17: c	Q18: c	Q20: a	Q21: a	Q22: c
Q23: b	Q24: b	Q25: a	Q26: b	Q27: a	Q28: c	Q29: a
Q30: a	Q31: a	Q32: b	Q33: b	Q34: a	Q35: b	Q36: b
Q37: a	Q38: a	Q39: b	Q40: a	Q41: a	Q42: b	Q43: b
Q44: b	Q45: b	Q46: c	Q47: c	Q48: b	Q49: b	Q50: b

Detailed answers

Q1. Answer: a `[Tuples]`

```
Output:
4
<class 'int'>
```

Why: True equals 1, so $1 + 3 = 4$, and the result type is int.

Q2. Answer: a `[Functions]`

```
Output:
True
```

Why: A function with no return statement returns None.

Q3. Answer: b `[f-strings]`

```
Output:
Mia: 87.6
```

Why: `:.1f` rounds the number to one decimal place.

Q4. Answer: b `[Tuples/Strings]`

```
Output:
123
15
```

Why: `"12" + "3"` concatenates to `"123"`; `int("12") + 3 = 15`.

Q5. Answer: b `[Operators]`

```
Output:
False
```

Why: $3*4=12$, $2+12=14$, $14 > 15$ is False, and False or False is False.

Q6. Answer: b `[Functions]`

```
Output:
9 8
```

Why: `power(3)=3**2=9`; keyword args allow reordering, so `2**3=8`.

Q7. Answer: b [\[Functions \(mutable default\)\]](#)

```
Output:  
[1]  
[1, 2]
```

Why: The default list is created once and persists between calls, so it accumulates.

Q8. Answer: c [\[Functions/Flow\]](#)

```
Output:  
not negative
```

Why: return exits the function immediately; the print after it never runs.

Q9. Answer: c [\[Lists \(mutation\)\]](#)

```
Output:  
3  
[5, 6, 1]
```

Why: append mutates the list to [5,6,1]; len is 3; the caller's list is changed.

Q10. Answer: a [\[Strings/Functions\]](#)

```
Output:  
ha-ha-ha
```

Why: [word]*3 makes three "ha", joined by "-".

Q11. Answer: c [\[Lists\]](#)

```
Output:  
[1, 2, 3]  
None
```

Why: append mutates the list in place and returns None.

Q12. Answer: a [\[Lists \(conv vs alias\)\]](#)

```
Output:  
[9, 2, 3]  
[9, 2, 3]  
[1, 8, 3]
```

Why: b is an alias of a (same list), so b[0]=9 changes a too; c is a real copy.

Q13. Answer: b [\[Comprehensions\]](#)

```
Output:  
[0, 9]
```

Why: range(6) keeps 0 and 3 ($x \% 3 == 0$); squared gives 0 and 9.

Q14. Answer: b [\[Lists\]](#)

```
Output:  
[3, 1, 2]  
[1, 2, 3]
```

Why: sorted() returns a new list; the original nums is unchanged.

Q15. Answer: b [\[Tuples/enumerate\]](#)

```
Output:  
1 a  
2 b  
3 c
```

Why: enumerate(start=1) makes the counter begin at 1.

Q16. Answer: b [\[Tuples/zip\]](#)

```
Output:  
[('a', 10), ('b', 20)]
```

Why: zip stops at the shortest iterable, so 'c' is dropped.

Q17. Answer: c [Dicts]

Output:
(prints 0, then raises KeyError)

Why: .get('b', 0) returns 0; dd['b'] then raises KeyError on the missing key.

Q18. Answer: c [Dicts]

Output:
7

Why: Iterating .items() and summing the values gives 2 + 5 = 7.

Q20. Answer: a [Tuples]

Output:
11

Why: "mississippi" has 11 characters, so the tuple has length 11.

Q21. Answer: a [Tuples (unpacking)]

Output:
1 [2, 3] 4

Why: Extended unpacking: first=1, middle=[2,3] (a list), last=4.

Q22. Answer: c [Dicts/Lists]

Output:
30

Why: record['scores'] is [10,20,30]; [-1] takes the last element.

Q23. Answer: b [Built-ins]

Output:
True
False

Why: any is True (2 and 3 are truthy); all is False because 0 is falsy.

Q24. Answer: b [Strings]

Output:
Stats 21

Why: strip() removes the spaces, then title() capitalizes each word.

Q25. Answer: a [NumPy (dtype)]

Output:
bool

Why: An array of only booleans has dtype bool.

Q26. Answer: b [NumPy (broadcasting)]

Output:
[11 12 13]

Why: The scalar 10 broadcasts to (is added to) every element.

Q27. Answer: a [NumPy (broadcasting)]

Output:
[[11 22 33]
[14 25 36]]

Why: The (3,) array matches the 3 columns and is added across both rows.

Q28. Answer: c [NumPy (broadcasting)]

Output:
(ValueError: shapes cannot broadcast)

Why: (2,3) + (2,) fails: 2 doesn't match the 3 columns on the right.

Q29. Answer: a `[NumPv (axis)]`

Output:
[3. 4.]

Why: axis=0 averages down each column: (1+3+5)/3=3, (2+4+6)/3=4.

Q30. Answer: a `[NumPv (nan)]`

Output:
[False True False]
2.0

Why: isnan flags the nan; nanmean ignores it, averaging 1 and 3 to 2.0.

Q31. Answer: a `[NumPv (where)]`

Output:
['small' 'big' 'small']

Why: where picks "big" where x>2, else "small".

Q32. Answer: b `[NumPv]`

Output:
[0.0, 0.5, 1.0]

Why: linspace includes both endpoints and returns 3 evenly spaced points.

Q33. Answer: b `[NumPv]`

Output:
[2 5 8]

Why: arange steps by 3 and stops before 10 (stop is exclusive).

Q34. Answer: a `[NumPv (fancy indexing)]`

Output:
[30 10 30]

Why: Fancy indexing picks positions 2, 0, 2 in that order (duplicates allowed).

Q35. Answer: b `[NumPv (views)]`

Output:
[1 0 0 4]

Why: y is a view of x[1:3]; setting y[:]=0 writes through to x.

Q36. Answer: b `[NumPv]`

Output:
[3 1 2]
[1 2 3]

Why: np.sort returns a new sorted array; x itself is unchanged.

Q37. Answer: a `[NumPv]`

Output:
1

Why: argmax returns the index of the first maximum (9 sits at index 1).

Q38. Answer: a `[NumPv (unique)]`

Output:
['A' 'B' 'C']
[3 2 1]

Why: unique returns sorted labels A,B,C with counts 3,2,1.

Q39. Answer: b `[NumPy (mask)]`

```
Output:  
[0 0 5 0]
```

Why: The boolean mask `x<0` selects the negatives and sets them to 0.

Q40. Answer: a `[NumPy]`

```
Output:  
[ True True]  
True
```

Why: `a==b` is element-wise (`[True True]`); `array_equal` returns one bool.

Q41. Answer: a `[Pandas / loc]`

```
Output:  
['a', 'b']  
20
```

Why: The dict keys become the index; `.loc['b']` returns 20.

Q42. Answer: b `[Pandas / shape]`

```
Output:  
(3, 2)
```

Why: The DataFrame has 3 rows and 2 columns.

Q43. Answer: b `[Pandas / boolean]`

```
Output:  
[True, True, False]
```

Why: `score >= 80` gives a boolean Series: 88 and 91 pass, 76 doesn't.

Q44. Answer: b `[Pandas / assign]`

```
Output:  
False  
True
```

Why: `assign` returns a new DataFrame; the original df is unchanged, so 'y' is only in `new_df`.

Q45. Answer: b `[Pandas / sort_values]`

```
Output:  
Noah
```

Why: Sorted by score descending, `iloc[0]` is the top row (Noah, 91).

Q46. Answer: c `[Pandas / value_counts]`

```
Output:  
3
```

Why: `value_counts` counts 'A' three times.

Q47. Answer: c `[Pandas / missing data]`

```
Output:  
2
```

Why: `isna().sum().sum()` totals the NaNs: one in x, one in y.

Q48. Answer: b `[Pandas / fillna]`

```
Output:  
[2.0, 3.0, 4.0]
```

Why: The mean of 2 and 4 is 3.0, which fills the NaN.

Q49. Answer: b `[Pandas / dropna]`

```
Output:  
2
```

Why: `dropna` removes the row with a NaN score, leaving 2 rows.

Q50. Answer: b [\[Pandas \(round\)\]](#)

Output:

15.0

Why: Group A's mean score is $(10 + 20) / 2 = 15.0$.